

# DeepSeek-V2

A Strong, Economical, and Efficient  
Mixture-of-Experts Language Model

**236B**

Total Parameters

**21B**

Activated / Token

**128K**

Context Length

**8.1T**

Training Tokens

Multi-Head Latent Attention • DeepSeekMoE

# The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

## The Problem

Scaling dense models improves quality but:

- Training costs grow super-linearly with parameter count
- KV cache size scales with model dimension  $\times$  layers, bottlenecking inference
- Generation throughput drops as models grow larger
- Existing fixes (GQA/MQA) trade performance for speed

## DeepSeek-V2's Answer

Two architectural innovations:

- **Multi-Head Latent Attention (MLA)**
  - Compresses KV cache by 93.3% while outperforming standard MHA
- **DeepSeekMoE**
  - Activates only 21B of 236B params, cutting training cost by 42.5%

## MLA + DeepSeekMoE in a Transformer Block

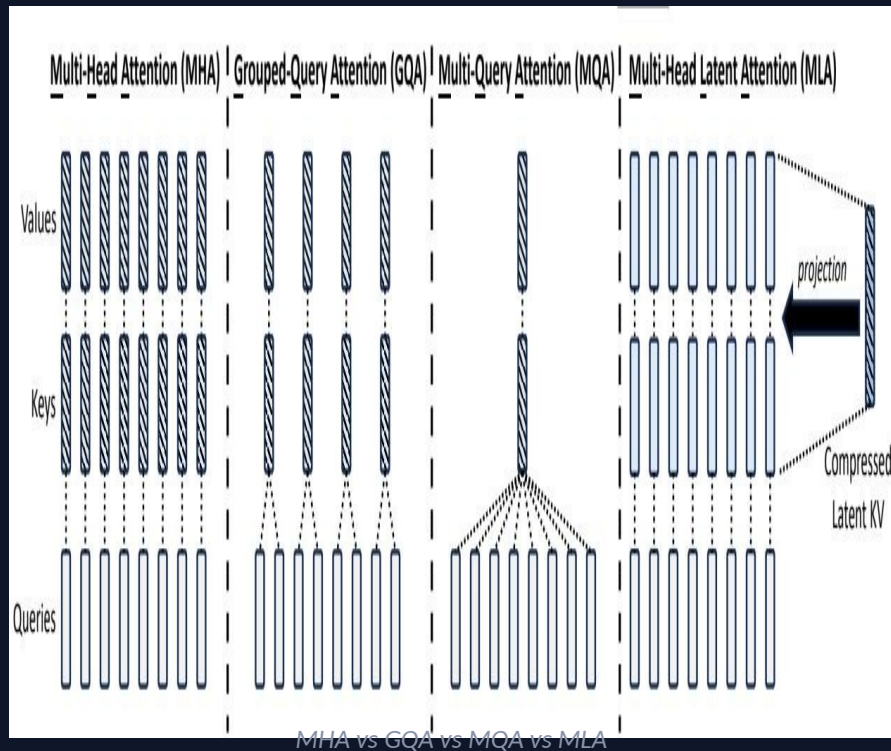


# Multi-Head Latent Attention (MLA)

Low-rank KV joint compression with decoupled RoPE

## How MLA Works

- Compresses K and V into a compact latent vector  $c_t^{KV}$  for caching
- Full K, V reconstructed on-the-fly via learned projections
- Decoupled RoPE-based positional key preserves position encoding
- **93.3% KV cache reduction vs. standard MHA**
- **Better accuracy than MHA, GQA, and MQA**



# DeepSeekMoE: Fine-Grained Experts

Shared expert isolation with device-limited routing



## Shared Experts

- Always activated for every token
- Capture common knowledge across inputs
- Reduce redundancy in routed experts



## Routed Experts

- Sparsely activated via top-K\_r gating
- Fine-grained segmentation increases specialization
- Only 21B of 236B params used per token



## Efficiency Design

- Device-limited routing: token hits  $\leq M$  devices
- Reduces cross-node communication overhead
- Auxiliary losses enforce load balance

*Token-dropping strategy during training prevents expert overflow without affecting inference quality*

# Efficiency Gains vs. DeepSeek 67B

Training cost, KV cache, and generation throughput improvements



42.5%

Training Cost  
Reduction



93.3%

KV Cache  
Reduction



5.76×

Generation  
Throughput

DeepSeek 67B (Dense)



67B activated · Baseline cost · 1× throughput

DeepSeek-V2 (MoE)



21B activated · 57.5% of cost · 5.76× throughput

# MMLU vs. Activated Parameters

DeepSeek-V2 achieves Pareto-optimal efficiency among open-source models

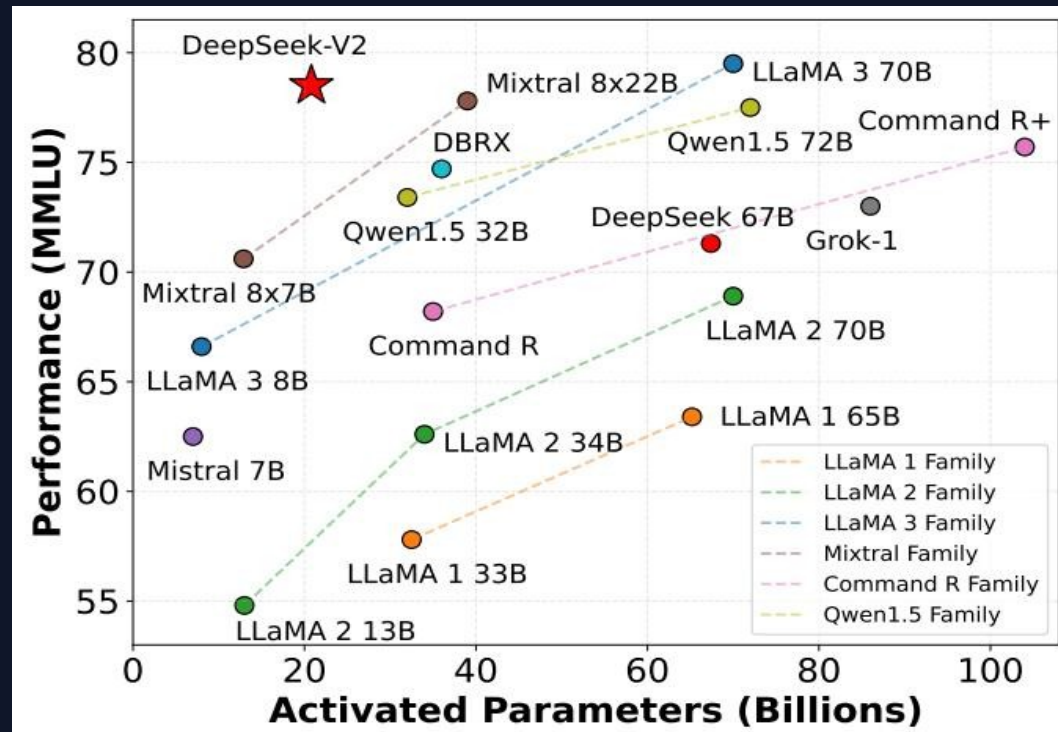


Figure 3: MMLU vs. Activated Parameters

## Pareto-Optimal

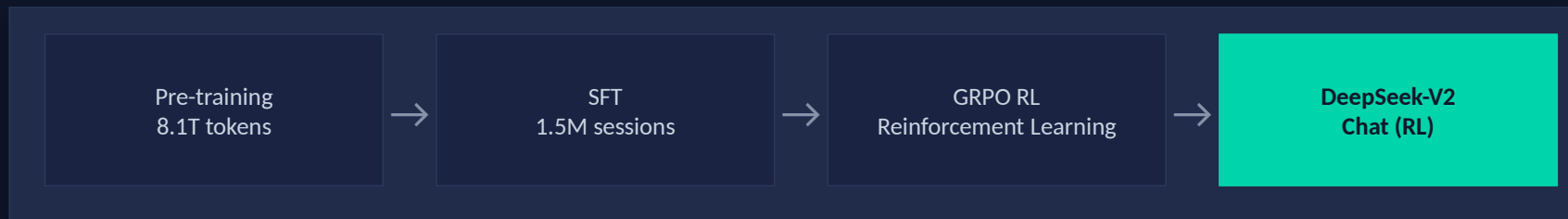
DeepSeek-V2 achieves the highest MMLU score (78.5) among all open-source models

Uses only 21B activated params — 3.3× fewer than LLaMA 3 70B

**Outperforms all 70B-class dense models on MMLU**

# Alignment & Chat Evaluation

SFT + GRPO reinforcement learning on 1.5M conversational sessions



AlpacaEval 2.0

38.9%

Length-controlled win rate

MT-Bench

8.97

Overall score

AlignBench

7.91

Overall (Chinese)

*DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source models on Chinese benchmarks*



# Key Takeaways

## Architectural Innovation Unlocks Efficient Scaling



### MLA > MHA for Inference

Strictly superior to GQA/MQA — 93.3% KV cache reduction with better accuracy



### Economical MoE Training

236B-parameter model trained at 42.5% lower cost than a 67B dense model



### Pareto-Optimal Performance

Only 21B activated params to match or exceed 70B-class dense models on MMLU



### Deployment-Ready Throughput

5.76× generation throughput improvement makes large-scale deployment practical



### Best Open-Source Chinese Chat

Competitive with closed-source systems in English; outperforms all open-source in Chinese